Layer 4: Gold Electrode Surface Characterisation

Roughness Factor, True Electroactive Area, Ohmic Budget, Area Variability, and Cleaning-State Shelf Life

Eton iGEM 2026, Dry Lab | Layer 4 of 17-Layer Biosensor Pipeline |
`electrode_characterisation.py`, `cleaning_protocol.py`

Layer 4 characterises the physical state of the gold working electrode (WE) surface that every downstream electrochemical layer treats as a fixed input. Unlike most of the pipeline it is not a dynamic simulation: it does not evolve a state variable in time (cleaning-state decay in Section IV is the one exception), but instead establishes four numbers — the true electroactive area A_{true}, the uncompensated resistance budget R_u, the inter-electrode area variability σ_A/A, and the post-clean contact angle θ — that Layers 6, 7, 9, 11, 13, and 14 all consume without re-deriving. Gold surface quality is one of the harder parameters to control in a disposable cartridge: batch-to-batch variation in roughness, contamination, and film thickness propagates directly into uncertainty on the surface coverage Γ that Layers 7–13 are built around, and is plausibly the single largest source of electrode-to-electrode variability in the device. The ratiometric design (Layer 13) cancels much, but not all, of this — Section V below checks exactly how much, using Layer 13's own equations rather than re-deriving them.

Scope note. This report supersedes no other layer: Layers 3 (mass transport) and 15 (calibration) are separately documented. It is built from the Layer 4 source material (`contents.md`, "Layer 4: Gold electrode surface") and supplies inputs to, but does not repeat, the surface-kinetics and ratiometric derivations of Layers 9 and 13.

I. Roughness Factor and True Electroactive Area

Definition and measurement

Evaporated gold on a disposable cartridge (thermally or e-beam evaporated Au on a Ti/Cr adhesion layer, on glass or PET/PETG, film thickness 50–200 nm) is never atomically flat. The electrochemical roughness factor R_f relates the geometric (mask-defined) area A_{geo} to the true, electroactive area:

$$R_f = \frac{Q_{\text{ox}}}{Q_{\text{ref}}\, A_{\text{geo}}}, \qquad A_{\text{true}} = R_f\, A_{\text{geo}} \tag{1}$$

where Q_{ox} is the charge under the Au-oxide-stripping peak, measured by cyclic voltammetry in 0.5 M H_2SO_4, and $Q_{\text{ref}} = 390\ \mu\text{C}\,\text{cm}^{-2}$ is the standard charge density for stripping one monolayer of Au oxide (Trasatti & Petrii 1991 [1] — a well-established literature constant, treated here as **confirmed**). R_f itself is **not** confirmed for this device: it must be measured per cartridge batch.

Representative literature bands: smooth evaporated gold, $R_f = 1.1$–1.5; annealed or polycrystalline gold, $R_f = 1.5$–3.0; rough electrodeposited nanostructured gold, $R_f \gg 3$. `electrode_characterisation.py` uses $R_{f,\text{nominal}} = 1.3$ as an illustrative placeholder for smooth evaporated Au, flagged `FLAG` throughout.

Why this matters for Γ

Every surface coverage Γ (mol cm^{-2}) used in Layers 7–13 must be expressed per unit *true* area, not geometric area:

$$\Gamma_{\text{true}} = \frac{\Gamma_{\text{geo}}}{R_f}, \qquad \Gamma_{\text{geo}} = \frac{Q_{\text{probe}}}{nF A_{\text{geo}}} \tag{2}$$

If R_f is not measured and is implicitly assumed to be 1 (i.e. $A_{\text{true}} \equiv A_{\text{geo}}$), Γ is overestimated by exactly $(R_f - 1)/1 \times 100\%$: for the nominal $R_f = 1.3$ this is a 30% error, consistent with the value quoted in the Layer 4 source material. We checked whether Layers 7 and 9 make this assumption explicit: both report Γ_p in pmol cm^{-2} without stating whether the denominator is A_{geo} or A_{true}. This report's convention — and the one Layers 7–13 should adopt — is that all Γ values are per true area, per Eq. (2); this should be confirmed explicitly in those layers' own code rather than left implicit.

Worked example

For $A_{\text{geo}} = 1\,\text{mm} \times 1\,\text{mm} = 0.01\,\text{cm}^2$ and a measured stripping charge $Q_{\text{ox}} = 4.5\,\mu\text{C}$:

$$R_f = \frac{4.5}{390 \times 0.01} = 1.15, \qquad A_{\text{true}} = 1.15 \times 0.01 = 0.0115\ \text{cm}^2$$

Figure 1A sweeps R_f over the full literature range and shows A_{true} against the three roughness bands above.

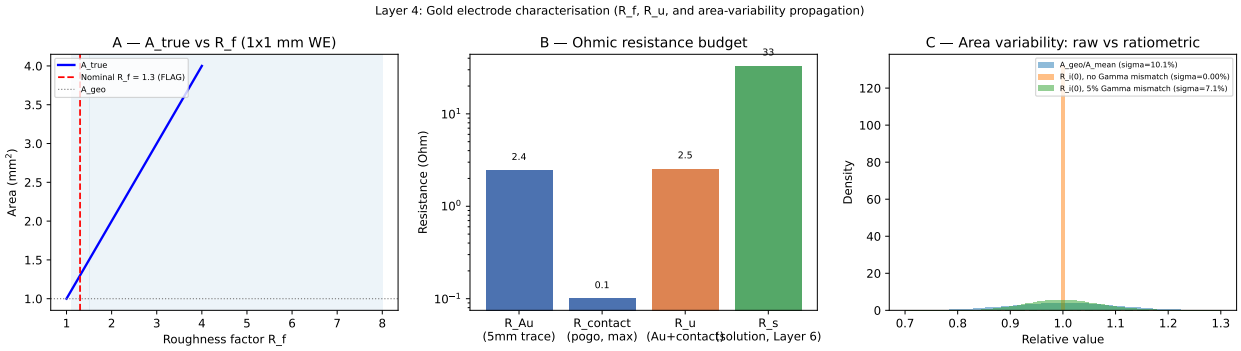


Figure 1: **A**: true area vs. roughness factor for a 1×1 mm WE, with the three literature roughness bands shaded and the nominal placeholder $R_f = 1.3$ marked. **B**: ohmic resistance budget (Section II) — gold-film and contact resistance are both $\ll$ the typical solution resistance R_s from Layer 6. **C**: Monte Carlo propagation of ±5–10% electrode-area variability (Section III) through to the raw electrode signal and to the ratiometric baseline $R_i(0)$, with and without a residual probe-coverage-ratio mismatch between MB and Fc electrodes.

II. Ohmic Resistance Budget

Electrical connection from bonding pad to potentiostat contributes three resistances in series with the solution resistance R_s that Layer 6 models:

$$R_{\text{Au}} = \frac{\rho_{\text{Au}}\, L_{\text{trace}}}{w\, t}, \qquad \rho_{\text{Au}} = 2.44 \times 10^{-8}\ \Omega\text{m (bulk gold resistivity)} \tag{3}$$

For a representative trace ($L = 5\,\text{mm}$, $w = 0.5\,\text{mm}$, $t = 100\,\text{nm}$): $R_{\text{Au}} = 2.44\,\Omega$ (Eq. (3), confirmed arithmetic). Contact resistance depends on the interconnect technology (Table 1): spring-pin pogo contacts and anisotropic conductive film (ACF) are both $< 0.1\,\Omega$ typical, wire bonds $< 0.01\,\Omega$. Combined, $R_{\text{Au}} + R_{\text{contact}} \lesssim 5\,\Omega$.

Contact technology	Typical range	Notes
Spring-pin pogo	10–100 mΩ	most practical for a disposable cartridge reader
ACF (anisotropic conductive film)	1–100 mΩ per contact	bonded, non-reusable
Wire bond (hybrid)	1–10 mΩ	lowest resistance, least disposable-compatible

Table 1: Contact-resistance ranges by interconnect technology (`electrode_characterisation.py`).

Against Layer 6's worked solution resistance ($R_s \approx 33\,\Omega$, for $\sigma \approx 15\,\text{mS/cm}$, $d_{\text{WE-RE}} = 5\,\text{mm}$, $A_{\text{cell}} = 1\,\text{cm}^2$), the film and contact terms add at most ∼8% to $R_u = R_{\text{Au}} + R_{\text{contact}} + R_s$ — R_s is confirmed as the dominant term, consistent with Layer 6. The resulting iR peak-potential shift, even at a worst-case $R_u \approx 35.5\,\Omega$ and SWV peak current $I_{\text{peak}} = 100\,\text{nA}$, is

$$\Delta V_{iR} = I_{\text{peak}} R_u = 100\,\text{nA} \times 35.5\,\Omega \approx 3.6\ \mu\text{V},$$

five to six orders of magnitude below the ∼65 mV SWV peak width (Layer 11), so **no iR compensation is required** at expected currents. This should be rechecked only if the electrode area — and hence peak current — is substantially enlarged from the $1\,\text{mm}^2$ base case.

III. Electrode Area Variability

Three independent fabrication tolerances contribute to inter-electrode variation in A_{geo} for a nominal 1×1 mm WE:

- Photolithography edge placement: $\pm 1\,\mu\text{m} \rightarrow \pm 0.2\%$ on area.
- Dicing / mask alignment: $\pm 0.1\,\text{mm} \rightarrow \pm 10\%$ on area (dominant term).
- Wet-etch undercut: ± 2–$5\,\mu\text{m}$ per side ($\pm 3.5\,\mu\text{m}$ mid-point, all 4 edges) $\rightarrow \pm 1.4\%$ on area.

Combined in quadrature: $\sigma_A/A = \sqrt{0.2\%^2 + 10\%^2 + 1.4\%^2} \approx 10.1\%$, at the upper end of the ±5–10% range in the Layer 4 source material, with dicing alignment dominating. A Monte Carlo sample of 2×10^5 electrodes (`sample_A_geo` in `electrode_characterisation.py`) reproduces this figure directly (Fig. 1C, blue).

Does the ratiometric design cancel this?

Layer 13 (Eq. I.1–I.3 of that report) shows the raw MB and Fc peak areas both scale linearly with the *same* $A^{(i)}_{\text{true}}$ on a given electrode i, and that $A^{(i)}_{\text{true}}$ cancels **exactly** in the ratio $R^{(i)} = A^{(i)}_{\text{MB}}/A^{(i)}_{\text{Fc}}$. We reproduce that cancellation numerically here: propagating the $\sigma_A/A = 10.1\%$ area distribution through to I_{MB}, I_{Fc}, and $R_i(0) = I_{\text{MB}}(0)/I_{\text{Fc}}(0)$ under the assumption that $\Gamma_{\text{MB}}/\Gamma_{\text{Fc}}$ is identical on every electrode gives $\sigma_{R_i(0)} \approx 0.0\%$ — full cancellation (Fig. 1C, orange spike). This is a genuine cross-layer confirmation: Layer 4's dominant variability source (electrode area) is, on Layer 13's

own model, invisible to the calibrated signal F_k, *provided* the MB and Fc probe densities co-vary with area in a fixed ratio.

That proviso is exactly the caveat the Layer 4 source material raises: if $\Gamma_{\mathrm{MB}}/\Gamma_{\mathrm{Fc}}$ itself varies between electrodes — e.g. because the two thiols compete slightly differently for surface sites at different local roughness — cancellation is only partial. Introducing an illustrative, area-independent 5% per-electrode mismatch between Γ_{MB} and Γ_{Fc} (function `propagate_to_ratiometric`, argument `gamma_mismatch_sigma`=0.05) leaves a residual $\sigma_{R_i(0)} \approx 7.1\%$ (Fig. 1C, green) — smaller than the raw 10.1% area variation, but not negligible against the ∼4.8% single-scan noise floor Layer 13 computes for σ_ρ/ρ. **This mismatch, not raw area variation, is therefore the area-related quantity worth measuring on a real batch of electrodes** (e.g. by SWV diagnostic comparing Γ_{MB} and Γ_{Fc} coverage independently per electrode), not σ_A/A itself.

IV. Surface Cleaning State and Shelf Life

Protocol selection

Manufacturing residues — photoresist, lamination adhesive, atmospheric hydrocarbons — must be removed before SAM formation, since thiol self-assembly is sensitive to organic contamination. Four candidate protocols are compared in Table 2; only two are compatible with an already-assembled, sealed disposable cartridge (piranha etch and electrochemical cycling both require direct acid contact with the electrode and cannot be performed post-assembly).

Protocol	$\theta_{\text{post-clean}}$	Compat. post-assembly?	Notes
O_2 plasma (2 min, 50 W)	0–10°	**Yes**	**chosen**: apply immediately before sealing
UV/ozone (15 min)	5–20°	Yes	milder fallback
Piranha (H_2SO_4/H_2O_2)	0–5°	No	incompatible with PET/PETG and a sealed cartridge
Electrochemical (CV, 20 cycles)	0–10°	No	needs acidic electrolyte in the cell

Table 2: Cleaning-protocol comparison (`cleaning_protocol.py`, `CLEANING_PROTOCOLS`).

O_2 plasma immediately before cartridge sealing is therefore the practical choice: post-plasma $\theta < 10°$, $R_f = 1.1$–1.3 for evaporated gold (no significant roughening from plasma exposure at these conditions).

Storage decay model

Contact angle is not stable in air: adventitious hydrocarbons re-adsorb over time, degrading SAM quality if the cartridge is assembled and stored before use. We model this as first-order recovery toward a contamination equilibrium:

$$\theta(t_{\text{storage}}) = \theta_0 + (\theta_\infty - \theta_0)\left(1 - e^{-k_{\text{decay}}\, t_{\text{storage}}}\right) \tag{4}$$

with $\theta_0 \approx 5°$ (post-plasma) and $\theta_\infty \approx 45°$ (air-equilibrated adventitious hydrocarbon layer). $k_{\mathbf{decay}}$ **is not measured for this device.** Ulman (Chem. Rev. 1996 [3]) supports hydrocarbon re-

adsorption on clean noble-metal surfaces occurring over hours to weeks, but gives no rate constant directly transferable to this geometry and packaging. Figure 2 sweeps three illustrative scenarios spanning that qualitative range ($\tau = 1/k_{\text{decay}} = 1$, 3, and 10 days) and reports the time to cross an illustrative SAM-quality threshold of $\theta = 30°$: 1.0, 2.9, and 9.8 days respectively (`shelf_life_days` in `cleaning_protocol.py`). The shelf-life estimate is essentially unconstrained without a real measurement — it varies by an order of magnitude across plausible k_{decay} values (Fig. 2B) — so this is flagged as a wet-lab priority rather than reported as a number.

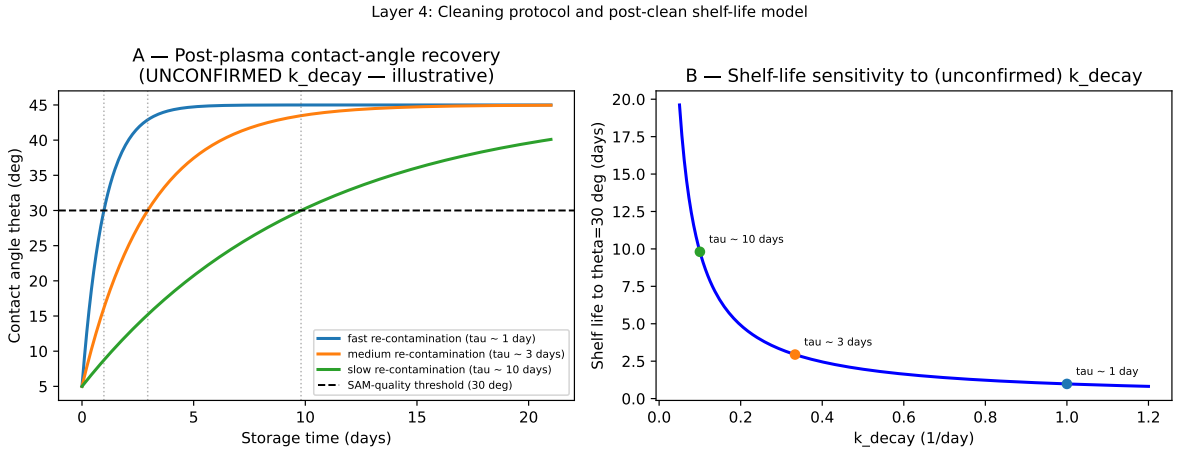


Figure 2: **A**: post-plasma contact-angle recovery under three illustrative, unconfirmed re-contamination rates, against a 30° SAM-quality threshold. **B**: shelf life to reach that threshold as a continuous function of k_{decay} — the near order-of-magnitude spread across the plausible range is the reason no single shelf-life number is reported as confirmed.

V. Grain Structure and Substrate Constraint

Evaporated gold on glass forms a polycrystalline, (111)-textured surface — the thermodynamically favoured face for thiol SAM formation. E-beam evaporation produces larger grains (10–50 nm) than thermal evaporation (∼5–10 nm); larger grains mean a smoother surface and lower R_f. Post-evaporation annealing (300°C, 20 min, vacuum or inert atmosphere) increases grain size further and reduces defect density, pushing R_f closer to 1 and giving more uniform SAM formation.

This has a direct, binary consequence for substrate choice: annealing at 300°C is compatible with a glass substrate but **not** with PET or PETG, which will deform well below that temperature. If the roughness-factor bands assumed elsewhere in this report (R_f = 1.1–1.5 for "smooth evaporated" gold, rather than 1.5–3.0 for unannealed polycrystalline gold) are to be achieved, the substrate decision effectively has to be made before the cleaning protocol in Section IV — a glass substrate keeps annealing available as an option; a PET/PETG substrate forecloses it and pushes the expected R_f toward the higher, unannealed band.

VI. Numerical Implementation

`electrode_characterisation.py`

Implements Eq. (1)–(2) (`rf_from_Q_oxide`, `A_true_cm2`, `gamma_true_from_gamma_geo`), the resistance budget of Section II (`R_Au`, `R_u_total`, `iR_shift`), and the area-variability Monte Carlo of Section III (`sample_A_geo`, `propagate_to_ratiometric`). Produces Figure 1.

`cleaning_protocol.py`

Encodes the protocol comparison of Table 2 (`CLEANING_PROTOCOLS`) and the storage-decay model of Eq. (4) (`theta_storage`, `shelf_life_days`), swept over the illustrative k_{decay} scenarios of Section IV. Produces Figure 2.

Both scripts follow the convention used throughout this pipeline: every literature or design constant that has not been measured on the fabricated device is named in a comment and printed at the `FLAG` / `UNCONFIRMED` level at runtime, so a wet-lab measurement can be substituted with a single-line change rather than a rewrite.

VII. Key Outputs

Output	Value (this report)	Status	Feeds
R_f	1.1–1.5 (evaporated Au)	**unconfirmed** — measure per batch	Layers 7–13 (Γ)
A_{true}	$A_{\text{geo}} \times R_f$	derived	Layers 7, 9, 11, 13
$R_{\text{Au}} + R_{\text{contact}}$	$< 5\,\Omega$ total	confirmed arithmetic	Layer 6 (R_u), Layer 14 (noise)
σ_A/A (raw)	$\pm 10.1\%$	derived from stated tolerances	cancels in $R_i(0)$ per Layer 13
Residual $\sigma_{R_i(0)}$ (5% Γ mismatch)	$\approx 7.1\%$	illustrative	Layer 13 (Sec. II uncertainty budget)
$\theta_{\text{post-clean}}$	$< 10°$ (O_2 plasma)	confirmed protocol choice	Layer 2 (capillary fill), Layer 7 (SAM)
Shelf life to $\theta = 30°$	1–10 days (scenario-dependent)	**unconfirmed**	packaging / expiry spec

Table 3: Layer 4 outputs and downstream consumers.

VIII. Limitations

1. R_f **is not measured.** Every A_{true}, Γ_{true}, and downstream figure in this report uses a placeholder $R_f = 1.3$. A single CV oxide-stripping scan per fabricated batch (Eq. (1)) resolves this and should be the first wet-lab check against this layer.
2. **Whether Layers 7 and 9 normalise Γ per true or geometric area is not stated in either report.** Section I.2 flags this; it should be confirmed explicitly (ideally by a one-line

comment or docstring in each layer's code) rather than left as an implicit assumption, since a silent mismatch reintroduces exactly the 30% error this layer exists to prevent.

3. **The 5% $\Gamma_{\mathbf{MB}}/\Gamma_{\mathbf{Fc}}$ mismatch used in Section III is illustrative, not measured.** It is the single most consequential unmeasured number in this report, since — unlike raw area variability — it does not cancel in the ratiometric signal. Measuring it requires comparing MB and Fc coverage independently per electrode (e.g. by separate SWV integration before probe cleavage), not just a total charge or area check.
4. **The storage-decay rate $k_{\mathbf{decay}}$ is unconstrained.** The 1–10 day shelf-life range in Section IV spans an order of magnitude and is not derived from device-specific data; Ulman (1996) supports only the qualitative existence and rough timescale of hydrocarbon re-adsorption, not a number for this packaging.
5. **Substrate choice (Section V) is not yet fixed**, and the R_f bands used throughout this report assume the more favourable, annealed-compatible case. If PET/PETG is chosen for cost or manufacturability reasons, R_f should be re-parametrised toward the 1.5–3.0 unannealed band, which would proportionally worsen every Γ-derived number in Layers 7–13.
6. **The area-variability model (Section III) treats the three tolerance sources as independent Gaussians at the stated 1-sigma widths.** The Layer 4 source material states them as ± ranges (more consistent with a uniform or manufacturer-spec interpretation); the Gaussian-RSS treatment here is a standard simplification but has not been validated against real inter-electrode measurements.

References

As cited in the Layer 4 source material (`contents.md`, "Key literature to find"); no local PDF is attached to this project for these four papers, so citations below are as given, not independently re-verified against the primary text.

[1] Trasatti S, Petrii OA (1991). Real surface area measurements in electrochemistry. *Pure and Applied Chemistry* 63(5):711–734. — source of $Q_{\text{ref}} = 390\ \mu\text{C}\,\text{cm}^{-2}$ (Section I).

[2] Hegner M, Wagner P, Semenza G (1993). Ultralarge atomically flat template-stripped Au surfaces for scanning probe microscopy. *Surface Science* 291(1-2):39–46. — evaporated gold surface characterisation, cited for grain-size and texture context (Section V).

[3] Ulman A (1996). Formation and structure of self-assembled monolayers. *Chemical Reviews* 96(4):1533–1554. — contact-angle recovery / hydrocarbon re-adsorption on clean metal surfaces (Section IV); qualitative support only, no device-specific rate.

[4] Bain CD, Whitesides GM (1988). Modeling organic surfaces with self-assembled monolayers. *Science* 240(4848):62–63. — SAM quality dependence on gold substrate quality, cited in support of the substrate/grain-structure discussion (Section V).